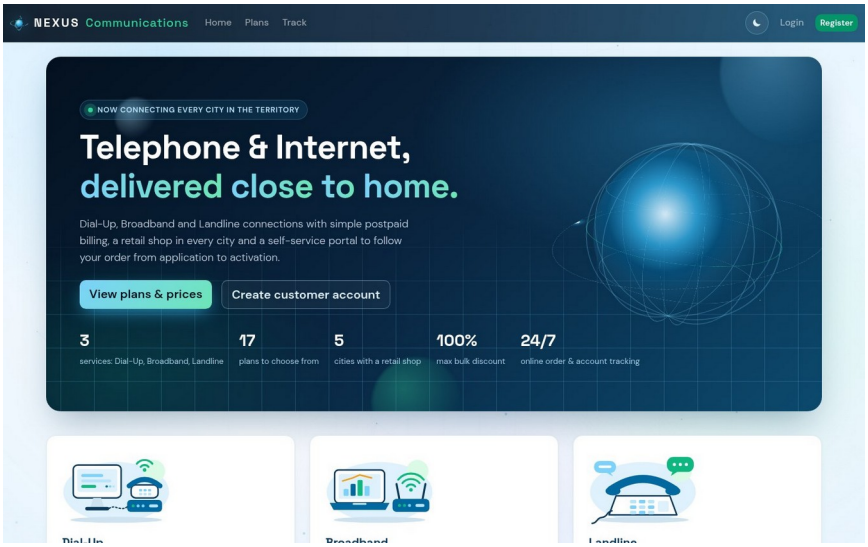
eProject · Project Report

Nexus Service Marketing System

Telephone & internet service management - orders, feasibility, connections, postpaid billing and retail operations for Nexus Communications



Item	Details
Project title	Nexus Service Marketing System
Technology	ASP.NET Core 8 MVC · Entity Framework Core 8 · SQL Server · Bootstrap 5
Project type	Group eProject - 5 members
Submission	October 2026

#	Team member	Student ID
1	**Ali Jan**	Student1661815
2	**Humaiz Khan**	Student1659420
3	**Muhammad Ehtisham Shakeel**	Student1657963
4	**Muhammad Waleed**	Student1659932
5	**Ali Turab**	Student1667727

Certificate of Completion

This is to certify that the project titled **"Nexus Service Marketing System"** has been designed, developed, tested and documented by the students named below as part of the eProject learning programme, under the guidance of the faculty mentor.

The project fulfils the problem statement issued for the eProject: it automates the record keeping of Nexus Communications for customers, orders, feasibility checks, connections, equipment stock, vendors, retail shops, employees, plans, postpaid billing, payments and customer feedback, through separate secure logins for each role.

Student name	Student ID	Signature
Ali Jan	Student1661815	
Humaiz Khan	Student1659420	
Muhammad Ehtisham Shakeel	Student1657963	
Muhammad Waleed	Student1659932	
Ali Turab	Student1667727	

Faculty mentor	Signature	Date

Project Team

Nexus Service Marketing System was built by a team of 5 students. Every member took part in analysis, testing and the final review; the table shows each member's main area.

#	Name	Student ID	Main responsibility
1	**Ali Jan**	Student1661815	Requirement analysis, admin module and project coordination
2	**Humaiz Khan**	Student1659420	Database design, ER diagram, entities and migrations
3	**Muhammad Ehtisham Shakeel**	Student1657963	Login and security, orders, feasibility and connections
4	**Muhammad Waleed**	Student1659932	Billing, payments, overdue control, tracking and search
5	**Ali Turab**	Student1667727	Interface design, customer documents, testing and documentation

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1. Problem Definition

1.1 Introduction

Nexus Communication is an organisation that provides telecommunication and internet services. It offers two types of internet connection - **Dial-Up** (which needs a Nexus landline) and **Broadband** - as well as **Landline** telephone connections. To meet growing demand it has opened retail shops in every city of its territory, where customers can enquire, place orders for new connections and pay their bills.

Before a connection is given, the company checks feasibility: the distance from the exchange and the availability of server capacity. Equipment such as modems and routers is bought from different manufacturing vendors, so stock must be tracked against the orders received.

1.2 Existing scenario

- Customer, order, connection, payment and vendor records are kept on paper, so slip-ups occur and searching is slow.
- Bulk and corporate schemes give 25% to 100% discount on the advance and security deposit depending on the number of connections, and are calculated by hand.
- Address proofs, application forms and receipts are filed manually by year and area.
- Bills, dues, replacement charges and discounts are worked out manually for every customer.

1.3 Proposed solution

An online application with a single SQL Server database and a separate login for each role - Admin, Accounts, Technical, Retail employee and Customer - that updates and maintains every record automatically:

- unique 11-character **order numbers** (D / B / T + 10-digit serial) and 16-character **account IDs** (type + 3-digit city code + 12-digit serial), generated by the system;
- search by order number or account ID, plus an **advanced search** by name, contact number, connection type and date or period;
- postpaid **billing** by the Accounts department with plan charges, call charges, deposits, discounts and **12.24% service tax**, with paid and due amounts always visible;
- **feasibility** checks, connection creation and activation status (active, temporarily inactive, permanently inactive) handled by Technical staff;
- customer self-service for contact details, orders, bills, documents and feedback.

1.4 Objectives

#	Objective	Achieved by
1	Replace paper records with one database	19 related tables in SQL Server, created by EF Core migrations
2	Give each role only the functions it needs	Cookie login, hashed passwords, role-based areas
3	Generate order numbers and account IDs automatically	IdentifierGenerator with atomic counters and CHECK constraints

#	Objective	Achieved by
4	Calculate bills exactly as the specification says	BillingService (plan validity, call rates, deposit, discount, tax)
5	Let anyone track an order or connection	Public Track page; amounts shown only to authorised users
6	Professional, responsive and accessible interface	Custom design system with motion, keyboard and reduced-motion support

2. Customer Requirement Specification

Every requirement of the problem statement, where it is implemented, and its status.

No.	Requirement (problem statement)	Implementation	Status
1	Separate logins for Admin, Accounts, Technical, Retail employees and Customers	Role areas, cookie authentication, hashed passwords	Done
2	Admin maintains employees, stock, vendors, retail shops and plans (insert, update, delete, search)	Admin area: Cities, Shops, Employees, Plans, Discounts, Vendors, Products, Purchases	Done
3	Order ID: 11 characters, D / T / B + serial	IdentifierGenerator + database CHECK constraint	Done
4	Account ID: type + 3-digit city code + 12-digit serial (16 characters)	IdentifierGenerator, Cities.Code	Done
5	Orders placed by retail employees (and by customers themselves)	Retail > Orders, Customer > Orders	Done
6	Feasibility: landline + internet when both applied for; internet only when the customer already has a Nexus landline	Dial-up order with landline plan; OrderWorkflowService	Done
7	Technical: update order status, create connections, temporary / permanent inactive, maintain equipment	Technical area: Feasibility, Connections, Overdue, Equipment	Done
8	Bills generated and updated only by Accounts; status of the connection depends on the bill	Accounts > Bills; Technical > Overdue suspends unpaid connections	Done
9	Payments by retail outlets and the Accounts office; paid and due amounts shown	Retail > Collect payment, Accounts > Record payment	Done
10	Bulk discount 25% / 50% / 75% / 100% on advance and security deposit	DiscountSchemes, counted over all the customer's connections	Done
11	Security deposits 325 / 500 / 250 $ and the full plan price list	Plans table (17 active plans + yearly STD awaiting price)	Done
12	Service tax 12.24% on the whole bill	BillingService	Done
13	Search order status by order ID and connection status by account ID	Public Track page	Done
14	Advanced search: ID, name, connection type, date or period, contact number	Admin > Advanced search, staff Search page	Done
15	Customer: contact details, place and track orders, check bills	Customer portal	Done

No.	Requirement (problem statement)	Implementation	Status
16	Address proofs, application forms and receipts filed by year and area	Customer Documents (PDF / PNG / JPEG, filed by year and city)	Done
17	Billing considers discounts, schemes and payments made previously	First-bill discount, balance brought forward on every bill	Done
18	Customer feedback collected	Customer > Feedback, Admin > Feedback	Done

3. Project Plan

The project followed seven phases. Each phase ended with a working, testable result.

Phase	Work	Output	Week
1. Analysis	Study the problem statement, roles and business rules	Requirement list (chapter 2)	1
2. Design	ER diagram, identifier formats, screen list, GUI standards	Database design, interface plan	1-2
3. Database	EF Core entities, configurations, CHECK constraints, seed data	19 tables, 5 migrations	2
4. Development	Five role areas and the business services	Working web application	2-3
5. Interface	Design system, 3D hero, motion, dashboards, responsive layout	site.css, site.js, views	3
6. Testing	Unit testing checklist, 48 automated end-to-end checks	Test results (chapter 11)	3-4
7. Delivery	Documentation, presentation, database backup, source ZIP	Final submission	4

4. System Design

4.1 Architecture

The application is a layered ASP.NET Core 8 MVC website. Each role has its own area protected by role-based authorization. Business rules live in services, not in controllers, so every screen applies the same rules. Entity Framework Core maps the entities to SQL Server.

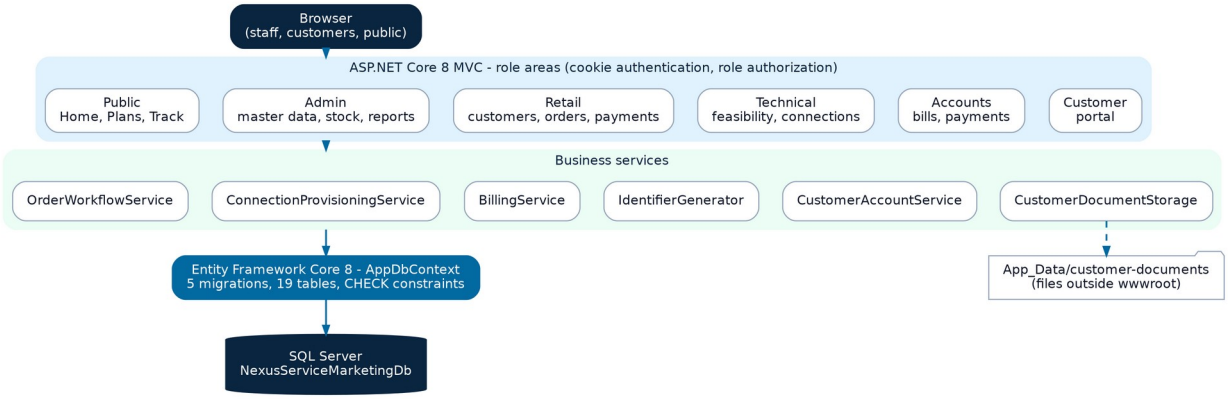


Figure 1 - System architecture

4.2 Roles and functions

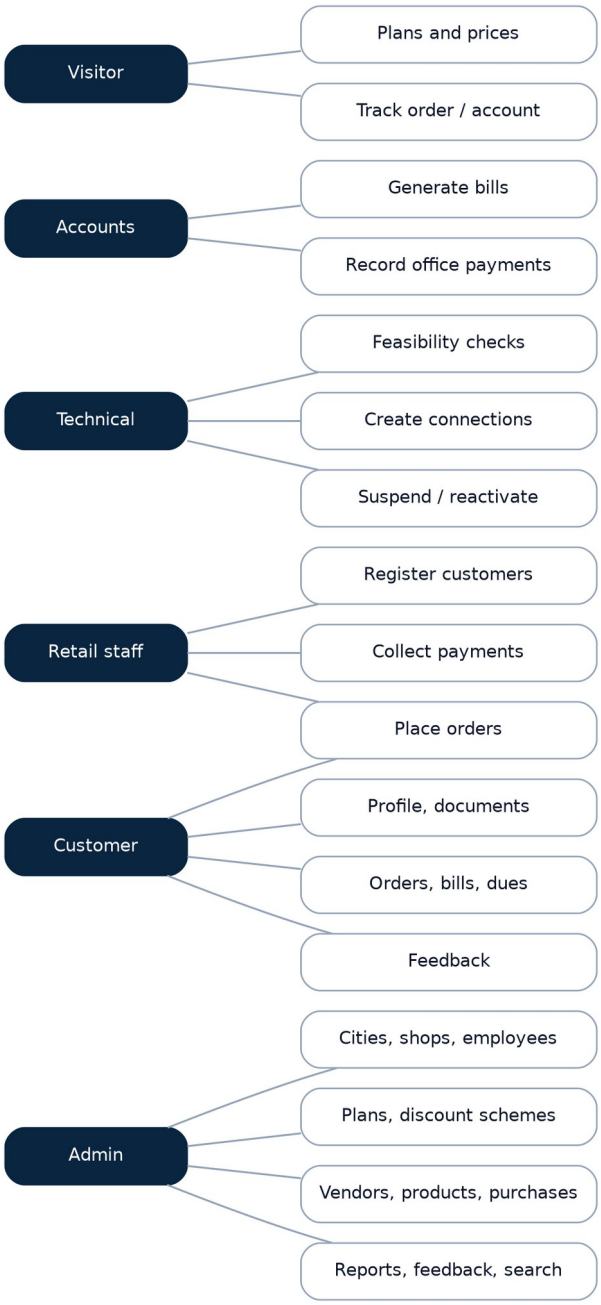


Figure 2 - What each role can do

4.3 Entity-Relationship Diagram

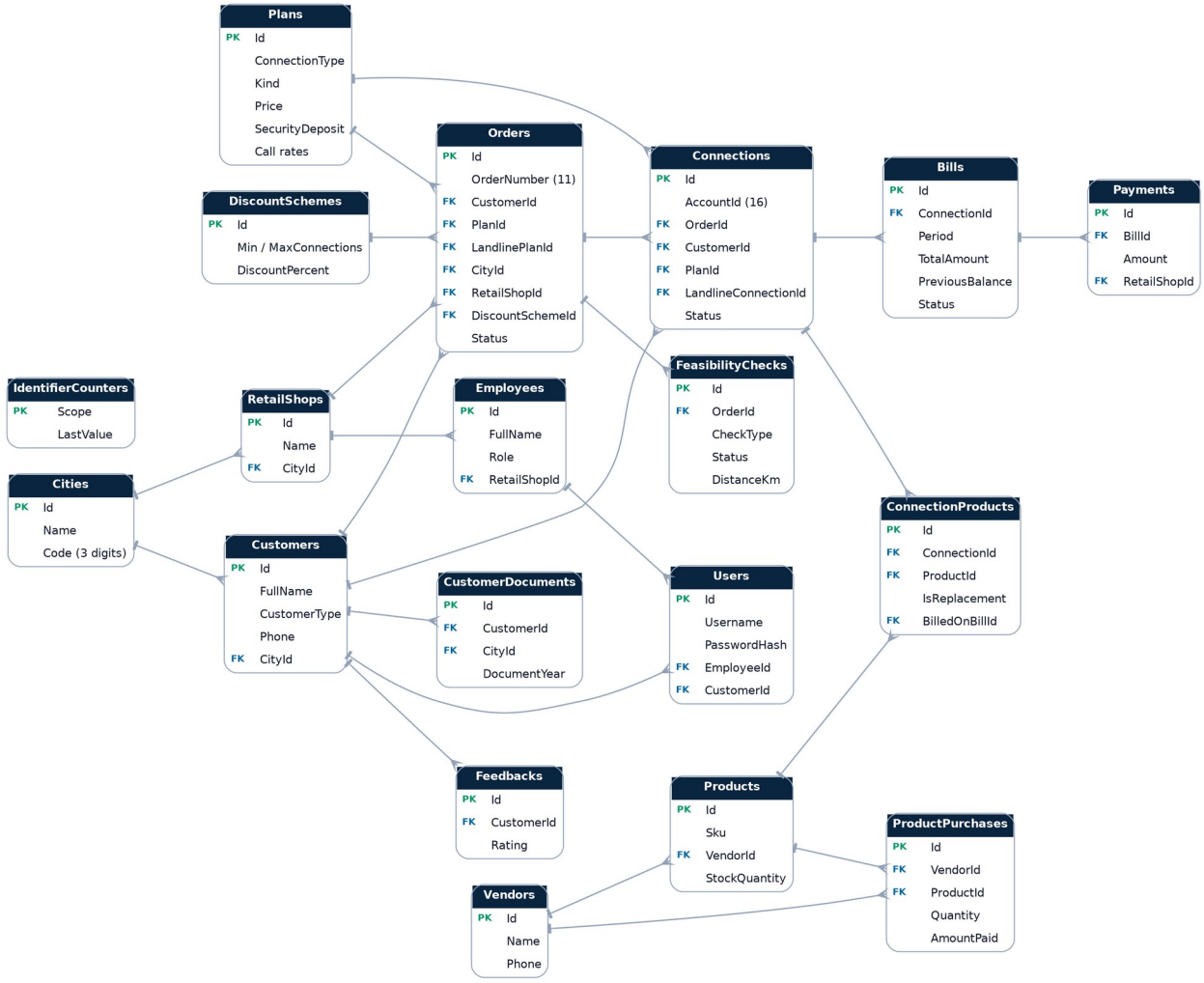


Figure 3 - Entity-relationship diagram (PK primary key, FK foreign key; crow's foot = many)

4.4 Data dictionary

Table	Purpose	Key rules
Users	Login accounts for staff and customers	Unique username, hashed password
Employees	Admin, accounts, technical and retail staff	Retail staff must belong to a shop
Customers	Personal and contact details	Individual or corporate
Cities	Cities of the territory	Unique 3-digit code used in account IDs
RetailShops	Retail outlets in each city	Unique name per city
Plans	Plan catalogue and charges	Hourly / unlimited / local / STD rules per kind
DiscountSchemes	Bulk discount bands	No overlapping bands at order time

Table	Purpose	Key rules
Orders	Applications for connections	Order number format D/B/T + 10 digits
FeasibilityChecks	Landline / internet feasibility results	Distance and server capacity
Connections	Connections provided	Account ID format; status values
ConnectionProducts	Equipment issued and replaced	Replacement charge billed once
Vendors, Products	Suppliers and equipment stock	Stock never negative
ProductPurchases	Purchases from vendors and amounts paid	Purchase increases stock
Bills	Postpaid bills	One bill per connection and period
Payments	Payments received	Never more than the amount due
Feedbacks	Customer feedback	Rating 1-5
CustomerDocuments	Filed proofs, forms and receipts	Filed by year and city
IdentifierCounters	Serial counters for IDs	Atomic increment under concurrency

4.5 Identifier formats

Identifier	Format	Example
Order number	Type letter (D / B / T) + 10-digit serial = 11 characters	D0000000001
Account ID	Type letter + 3-digit city code + 12-digit serial = 16 characters	B001000000000001

5. Algorithms

5.1 Order placement

1. Retail employee (or the customer) chooses customer, service type, plan, city, address and quantity.
2. The plan must be active and of the chosen service type; the city must be active.
3. Dial-up: if the customer has no active Nexus landline a **landline plan is required** - the telephone line and dial-up are applied for together.
4. Bulk band: count the customer's connections (not permanently closed) + open orders + this order; attach the matching discount scheme.
5. Take the next serial for the type atomically and build the order number; status = Placed.

5.2 Feasibility check

1. Technical starts the checks: Telephone - landline check; Broadband - internet check; Dial-up - both checks when a line is applied for, otherwise internet only.
2. For each check record the distance and, for internet, server capacity.
3. Any check not feasible - order Not feasible. All feasible - order Feasible.

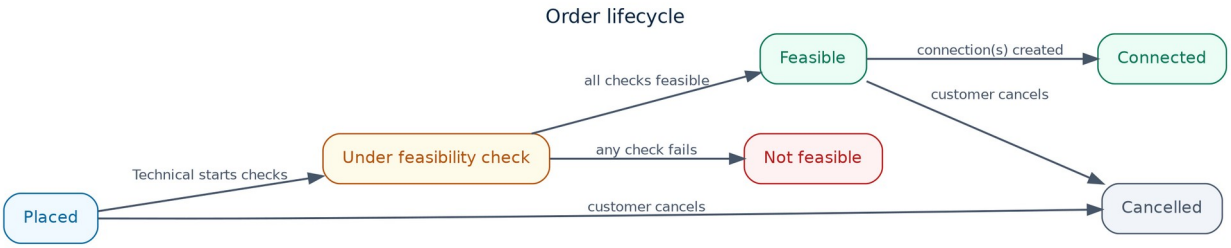


Figure 4 - Order lifecycle

5.3 Connection creation

1. Order must be Feasible. Internet orders take one modem / router per connection from stock.
2. Deposit = plan deposit, reduced by the bulk discount when the scheme covers deposits.
3. For a bundled dial-up order, first create each telephone line (with its own number), then the dial-up connection linked to it.
4. Account ID = type + city code + next 12-digit serial for that type and city. Status Active; order Connected.

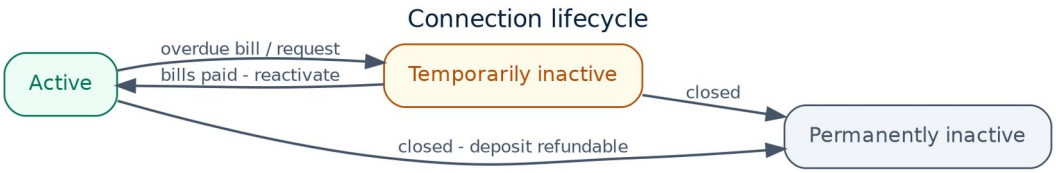


Figure 5 - Connection lifecycle

5.4 Bill generation

1. Plan fee: charged only when the period starts after the date the current plan is already paid to (plan validity).
2. Call charges = local x local rate + STD x STD rate + mobile x mobile rate (landline plans only).
3. Add unbilled equipment replacement charges and mark them billed.
4. First bill only: add the security deposit and subtract the bulk discount on the advance.
5. SubTotal = plan fee + usage + deposit + replacements - discount; tax = 12.24% of SubTotal; Total = SubTotal + tax.
6. Record the unpaid balance of earlier bills as **brought forward**.

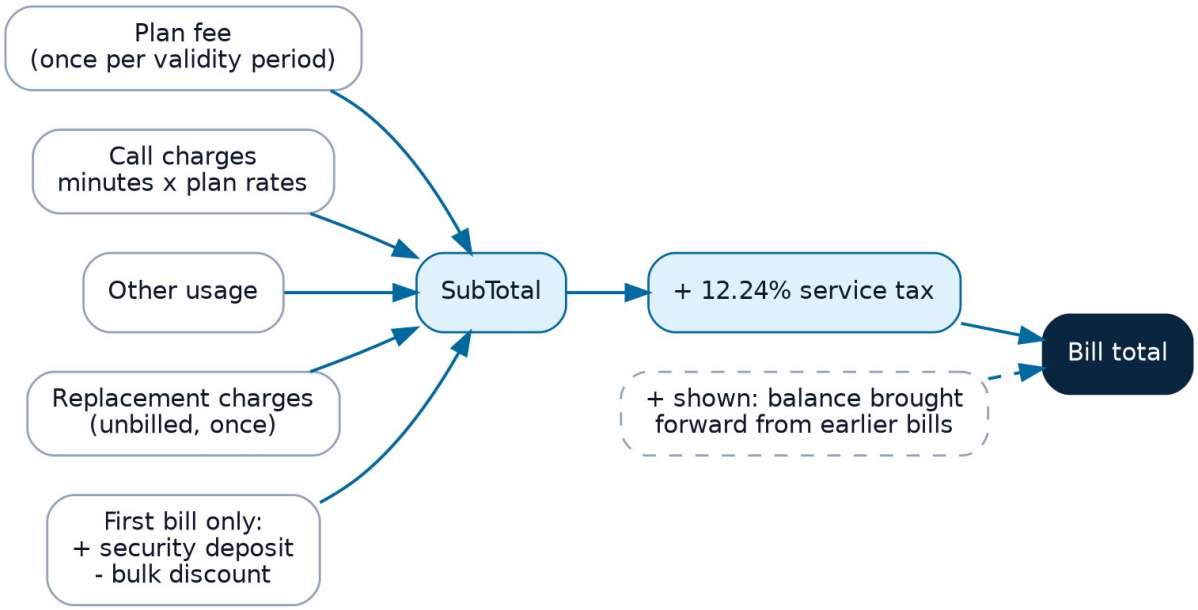


Figure 6 - How a bill is built

> **Worked example (verified by the automated tests)**
>
> Landline Local Monthly plan, first bill, 100 local minutes at $0.75:
>
> Plan fee $35.00 + calls $75.00 + deposit $250.00 = SubTotal $360.00
>
> Service tax 12.24% = $44.06 · Bill total = **$404.06**

5.5 Payments and overdue connections

1. A payment can never exceed what is still due; the bill becomes Partially paid or Paid.
2. Retail staff collect only bills of orders handled by their shop or self-service orders.
3. An active connection with a bill past its due date and not fully paid appears in Technical > Overdue and can be set temporarily inactive; once every bill is paid it is listed for reactivation.

6. GUI Standards

The interface follows one design system, defined as tokens in wwwroot/css/site.css, so every screen looks and behaves the same. Users can switch between a **light** and a **dark** theme with the sun / moon button in the top bar; the choice is remembered on that computer. Work surfaces are solid cards, so text stays sharp on any screen or projector.

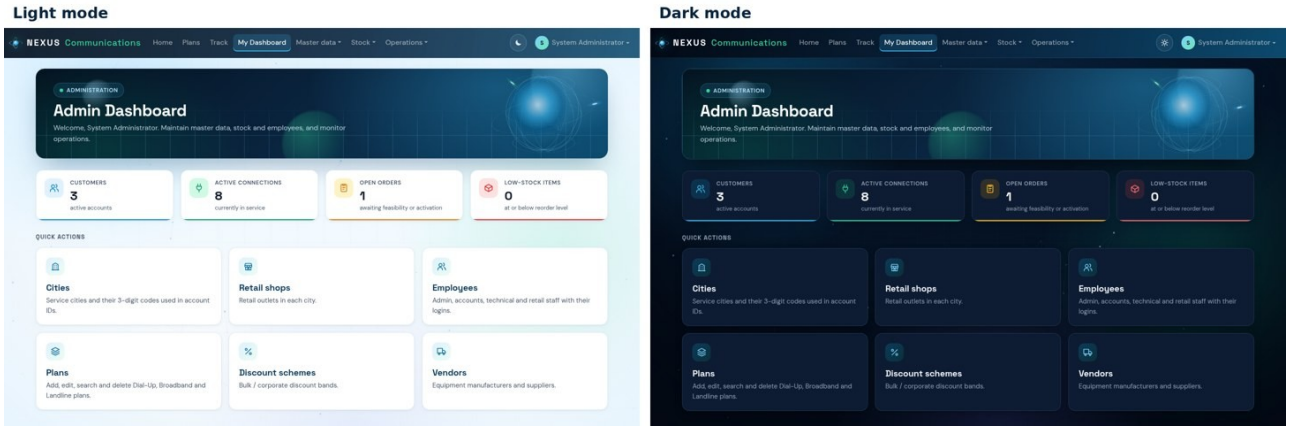


Figure 7 - The admin dashboard in light mode (left) and dark mode (right)

6.1 Colour palette

Swatch	Role	Light theme	Dark theme	Use
	Page background	#F0F9FF	#050F1E	Canvas behind the animated background
	Card surface	#FFFFFF	#0D1C32	Cards, tables, forms (solid)
	Text	#0F172A	#E2E8F0	Body text (contrast above 12:1)
	Primary	#0369A1	#0369A1	Buttons, links, active states (white text 5.9:1)
	Accent	#059669	#34D399	Success, call to action
	Danger	#DC2626	#F87171	Errors, overdue
	Navy stage	#0A2540	#0A2540	Top bar, hero banner, footer (both themes)

6.2 Typography and layout

Element	Standard
Headings	Space Grotesk, 600-700 weight (bundled with the app, works offline)
Body text	DM Sans 16 px, line height 1.55
Grid	Bootstrap 5 responsive grid; tested at 390, 768, 1024 and 1440 px
Navigation	Sticky navy bar, role menu, theme switch, account drop-down; current page highlighted
Page header	Title bar with breadcrumb on inner pages; 3D stage with globe on home, login and dashboards

Element	Standard
Forms	Labels above fields, validation next to the field, submit button shows a spinner
Tables	Uppercase headers, row hover, coloured status badges, scroll sideways on phones
Money	US dollars, two decimals

6.3 Illustrations

Ten original drawings were made for the project (wwwroot/images/illustrations.svg) - no stock photos, so there are no copyright issues and pages stay fast. Their colours come from CSS variables, so each drawing re-colours itself for the light and dark theme.

Page	Illustrations
Home - services	Computer with modem and phone (Dial-Up), laptop with Wi-Fi router (Broadband), desk phone (Landline)
Home - sections	Nexus retail shop, four "How it works" steps (order, feasibility, connection, bill), city skyline with a telecom tower
Plans	The matching service drawing beside each plan group
Register	Customer ID card with a check mark

6.4 Motion and accessibility

- Animated page background: slowly drifting colour glows, a moving grid and a particle network that reacts to the mouse (colours follow the theme).
- Loader on the first page of a visit; page cross-fade with a top progress bar.
- 3D globe and parallax layers on the hero, tilt and glare on cards, magnetic and ripple buttons.
- Scroll reveal and count-up figures; all motion uses transform and opacity only.
- Every animation is switched off for users who choose reduced motion; hover effects are skipped on touch screens.
- Visible keyboard focus, skip-to-content link, text contrast of at least 4.5:1 in both themes, labels on every field.

7. Interface Design Document

Screens of the finished application, taken from the running system with sample data.

Role	Main screens
Public	Home, Plans and prices, Track order / account, Login, Register
Admin	Dashboard, Cities, Retail shops, Employees, Plans, Discount schemes, Vendors, Products, Purchases, Customers, Orders, Advanced search, Reports, Feedback
Retail	Dashboard, Customers, Place order, Orders, Connections, Collect payment, Payment history, Search
Technical	Dashboard, Feasibility, Connections (create, status, plan, replace equipment), Overdue, Equipment, Search
Accounts	Dashboard, Generate bill, Bills and payments, Search
Customer	Dashboard, Profile, Orders (place, track, cancel), Connections, Bills, Documents, Feedback

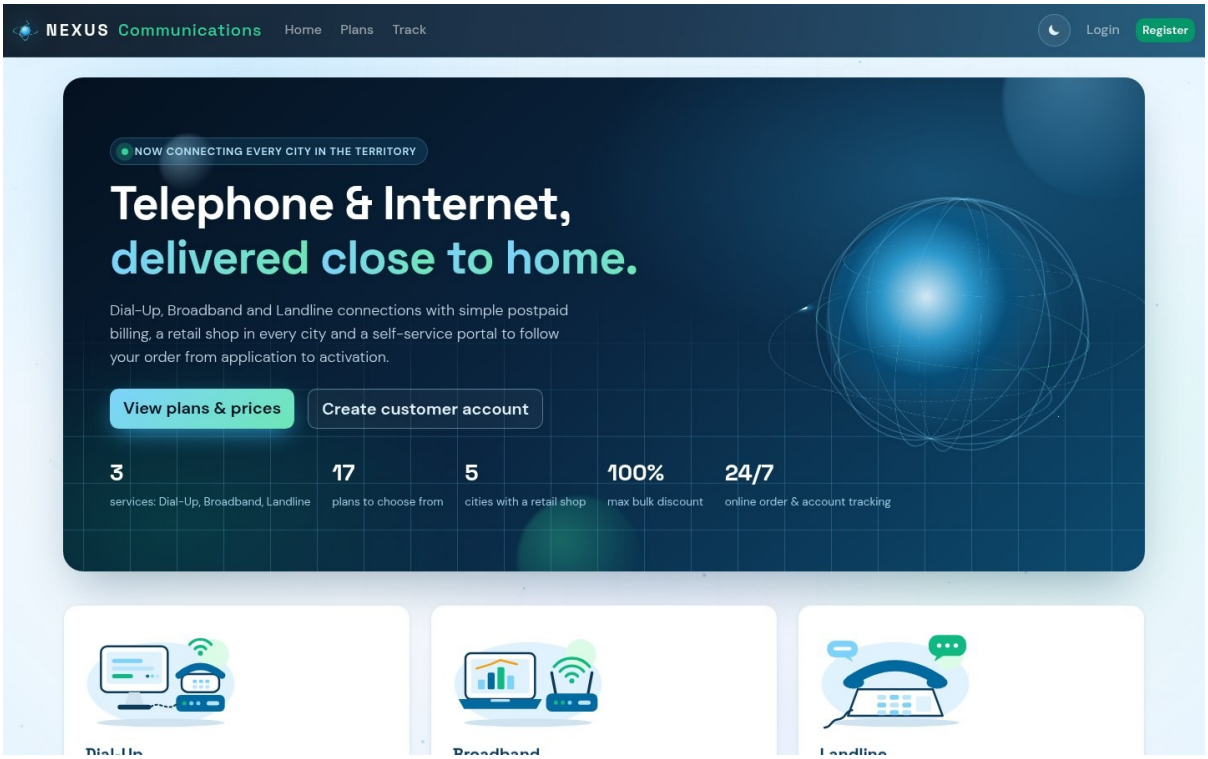


Figure 8 - Home page - 3D parallax hero, live figures and services

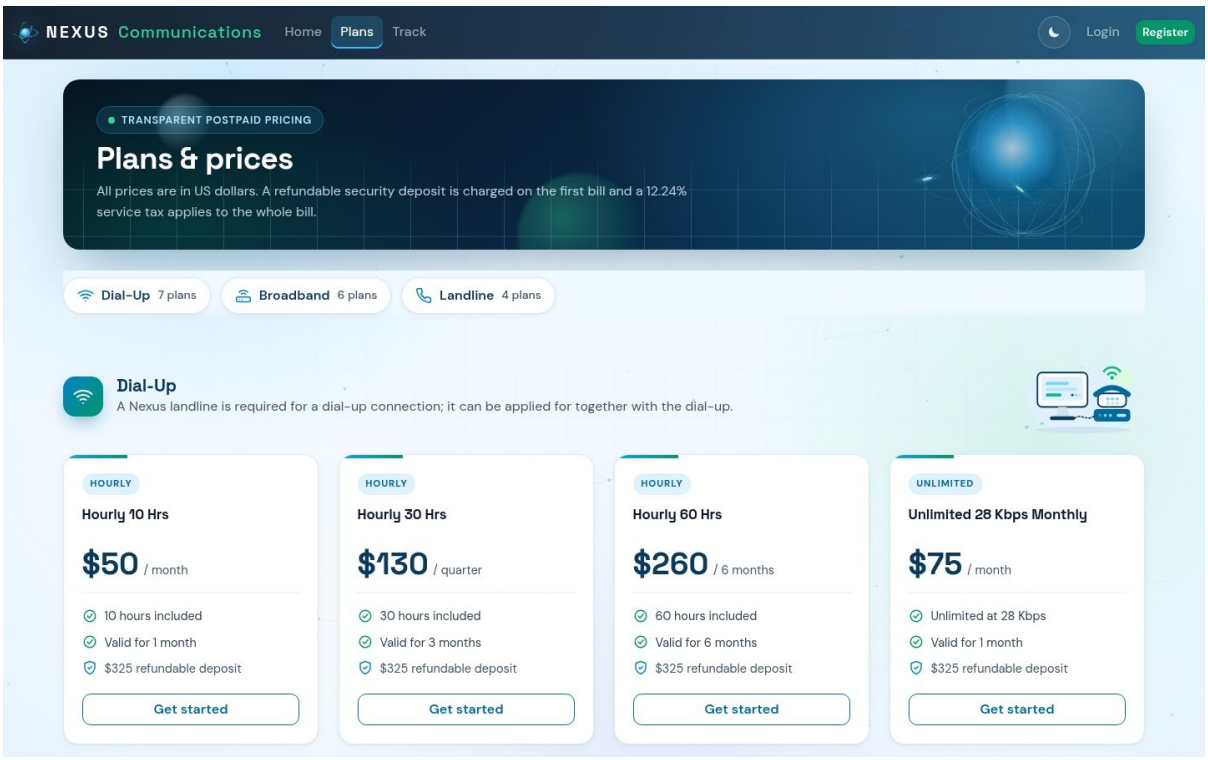


Figure 9 - Plans and prices - pricing cards grouped by service

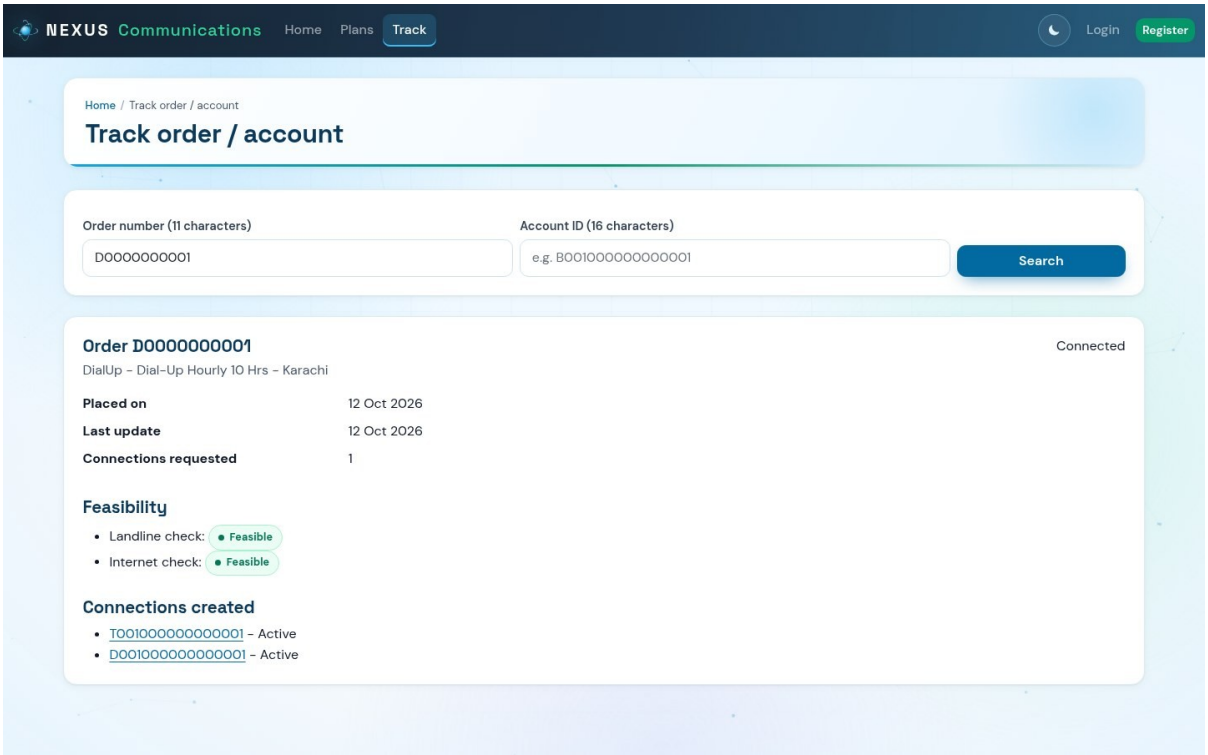


Figure 10 - Public tracking of an order number

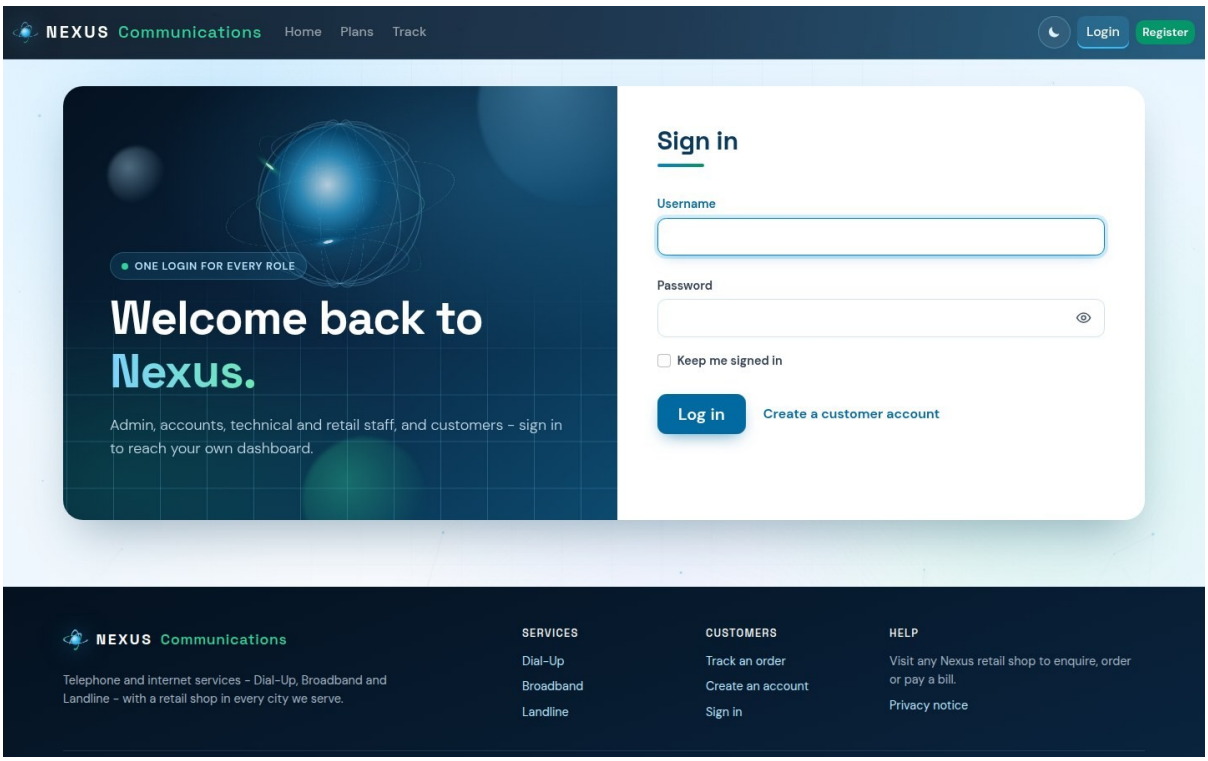


Figure 11 - Sign in - one login page for every role

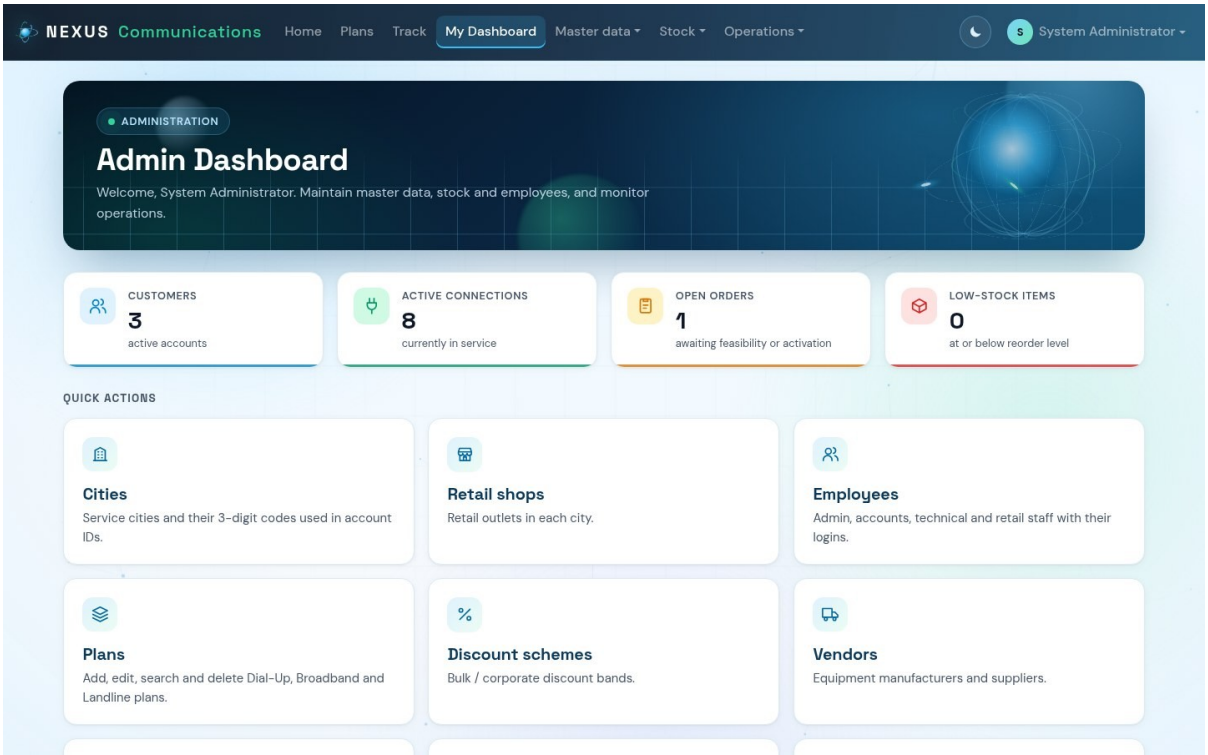


Figure 12 - Admin dashboard - live key figures and quick actions

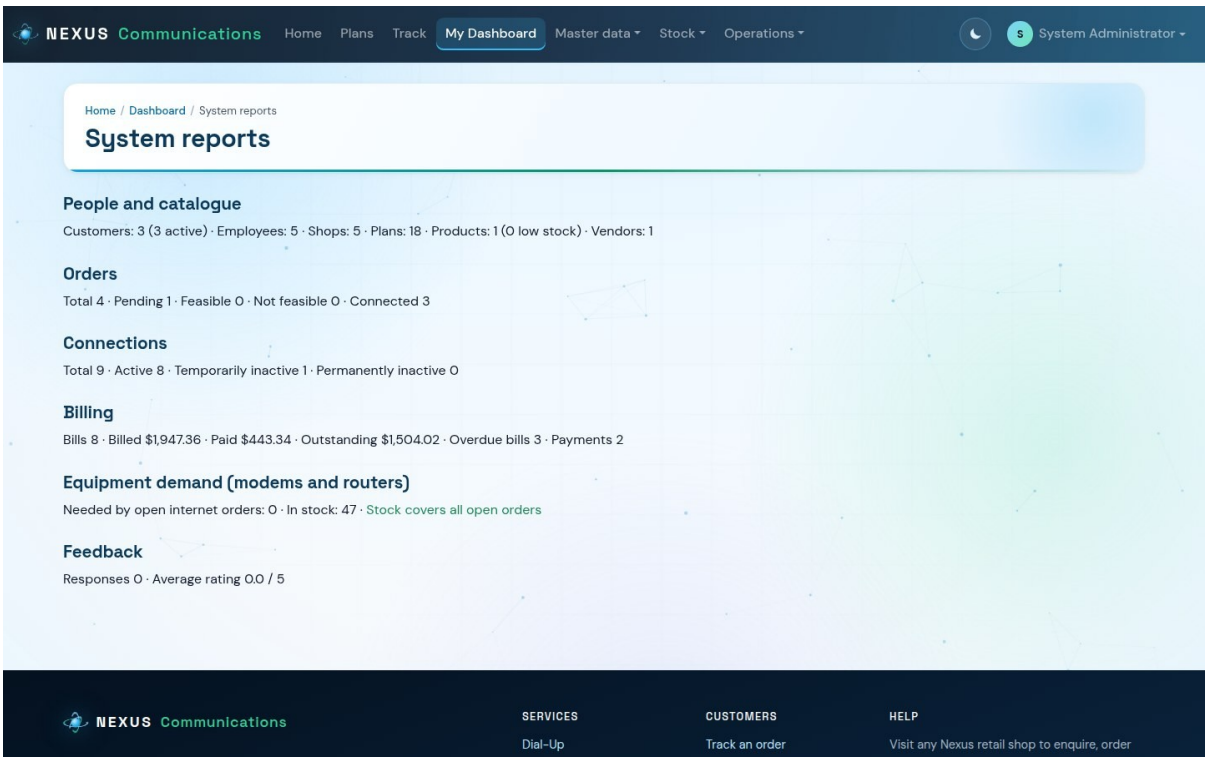


Figure 13 - Admin reports - orders, connections, billing and equipment demand

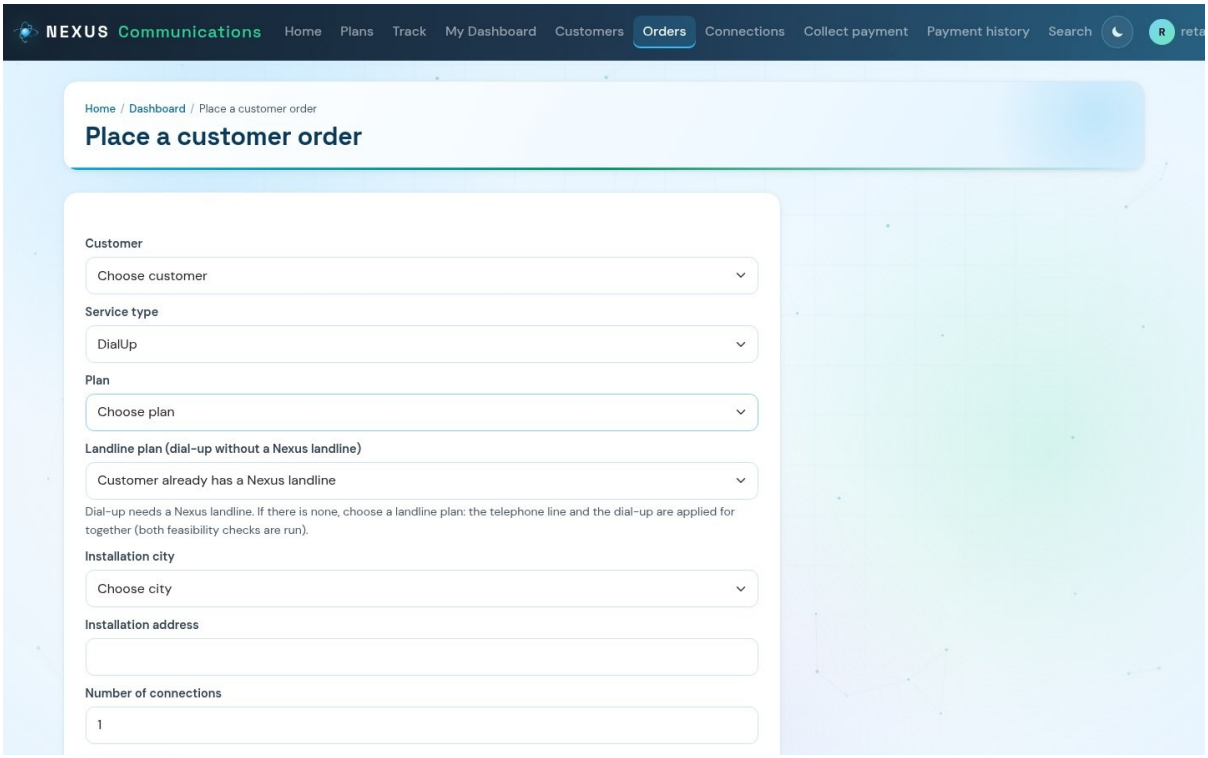


Figure 14 - Retail - placing an order (dial-up with a new landline)

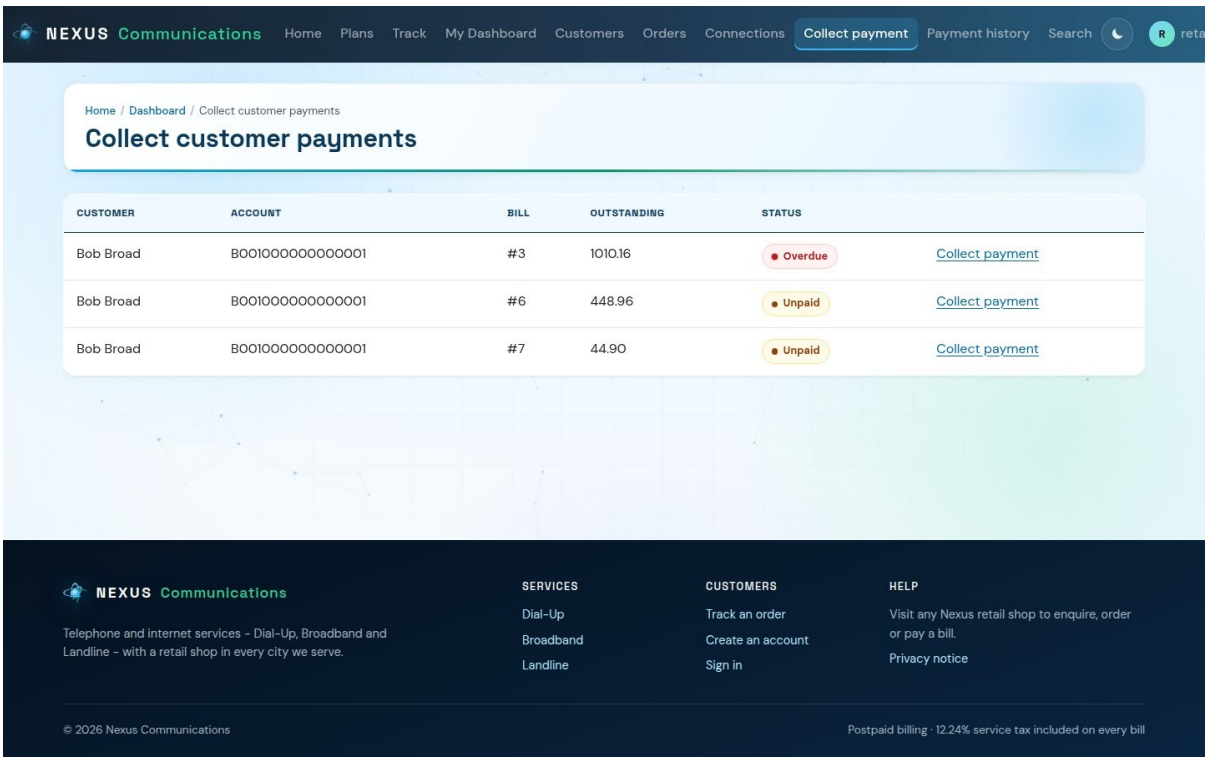


Figure 15 - Retail - bills the shop can collect

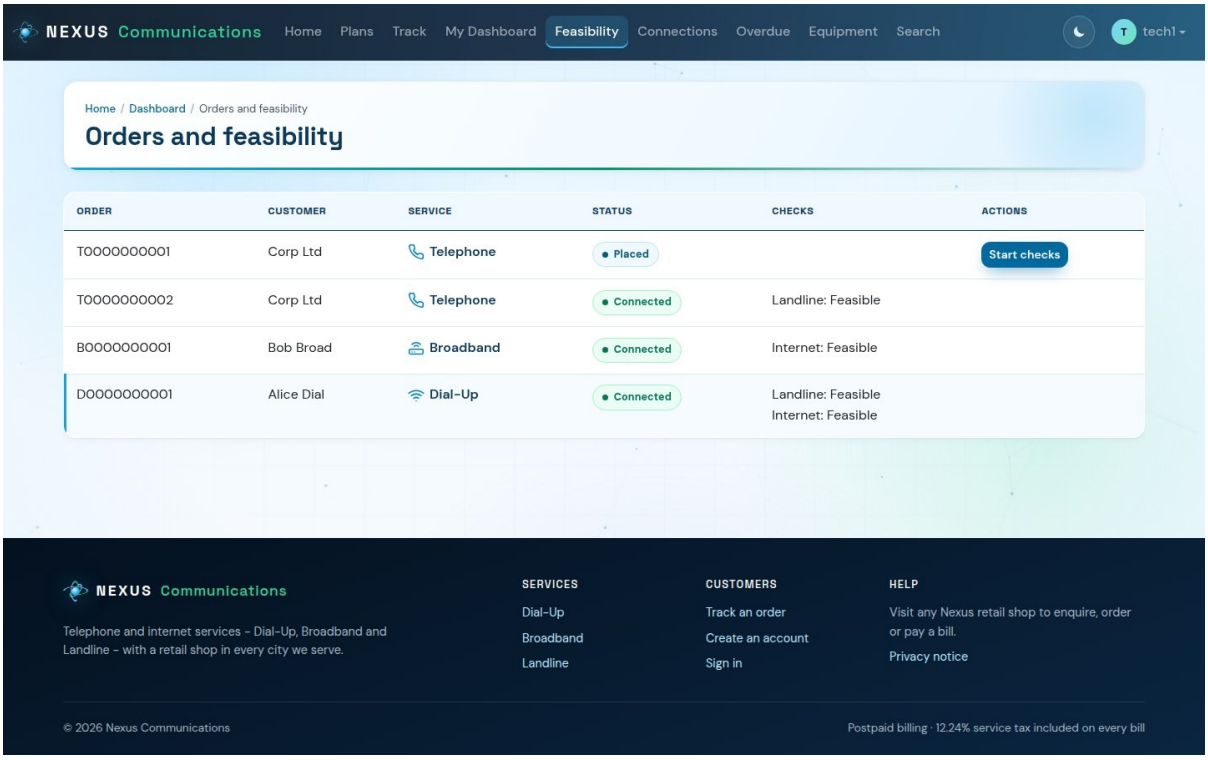


Figure 16 - Technical - feasibility queue

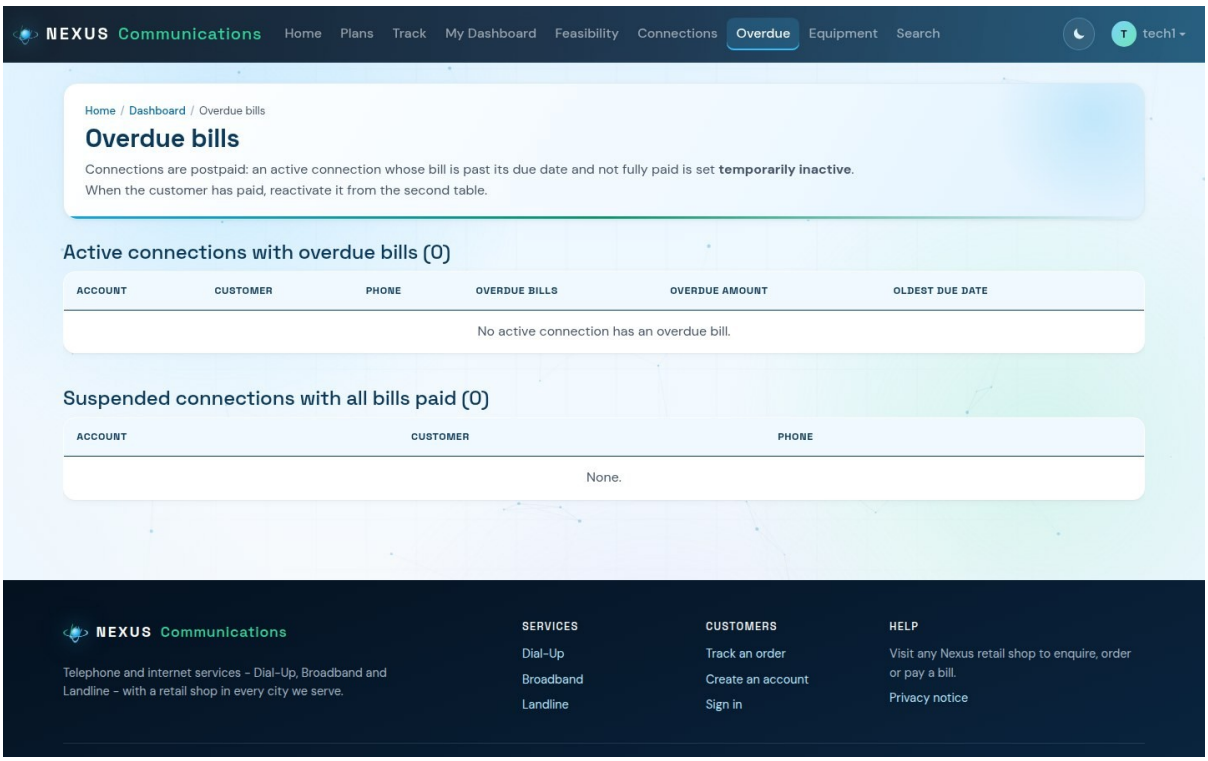


Figure 17 - Technical - overdue bills: suspend and reactivate

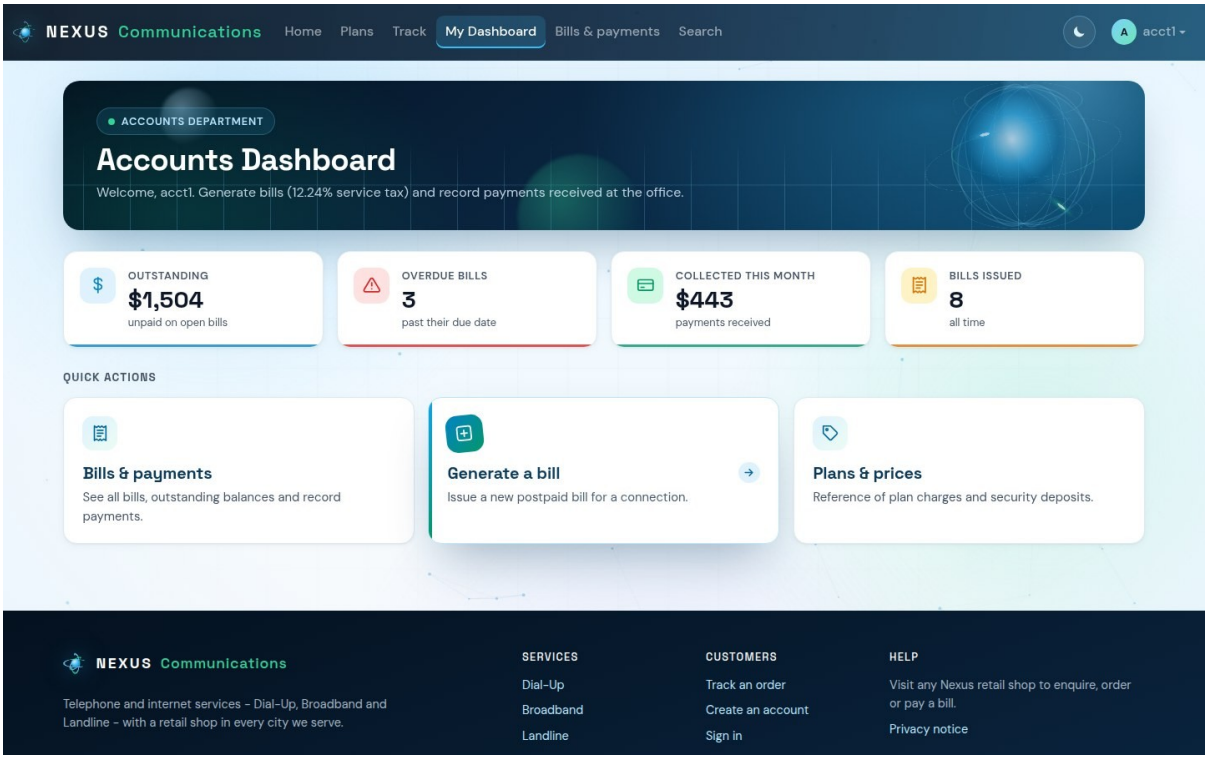


Figure 18 - Accounts dashboard

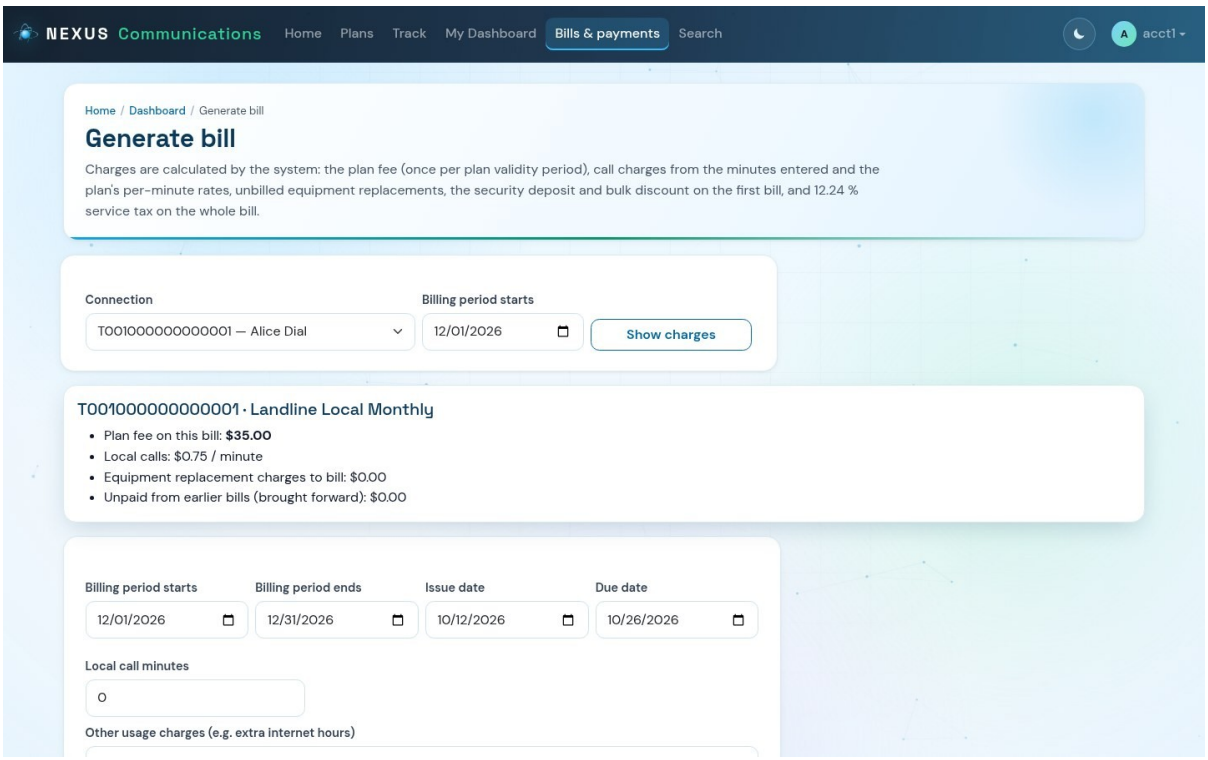


Figure 19 - Accounts - generating a bill with the charges preview

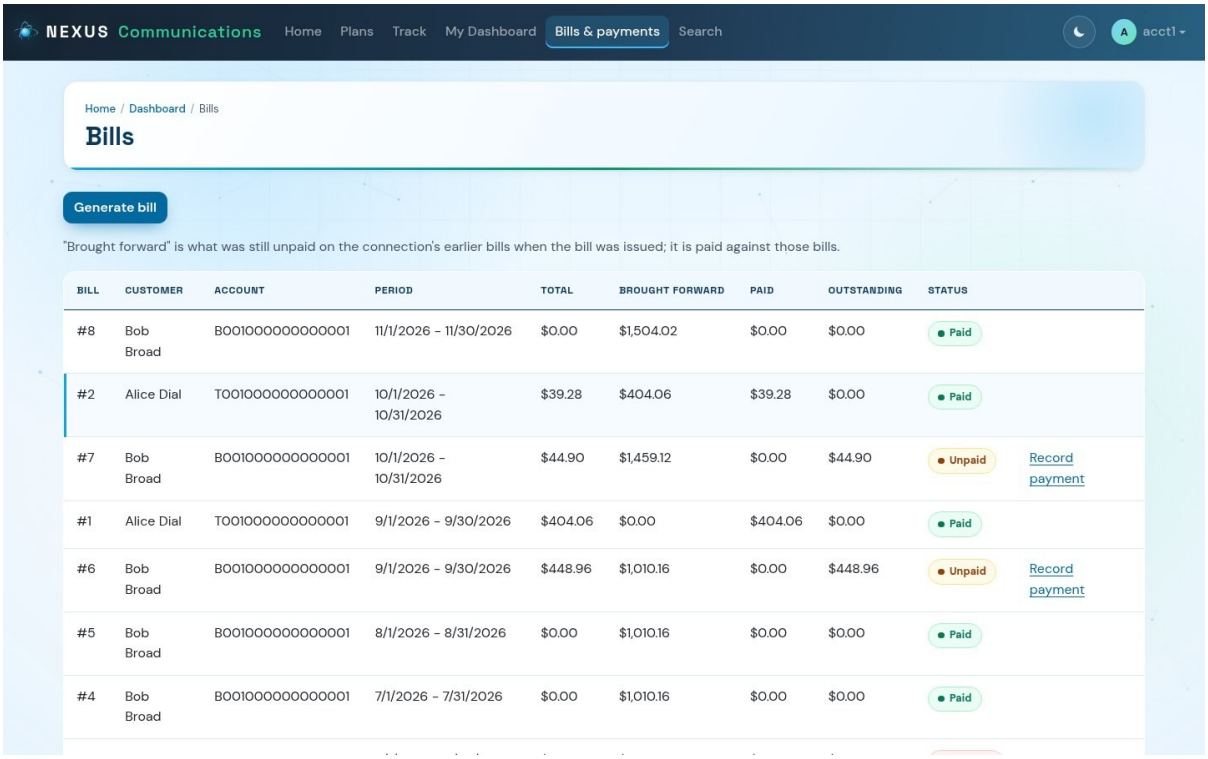


Figure 20 - Accounts - all bills with status badges

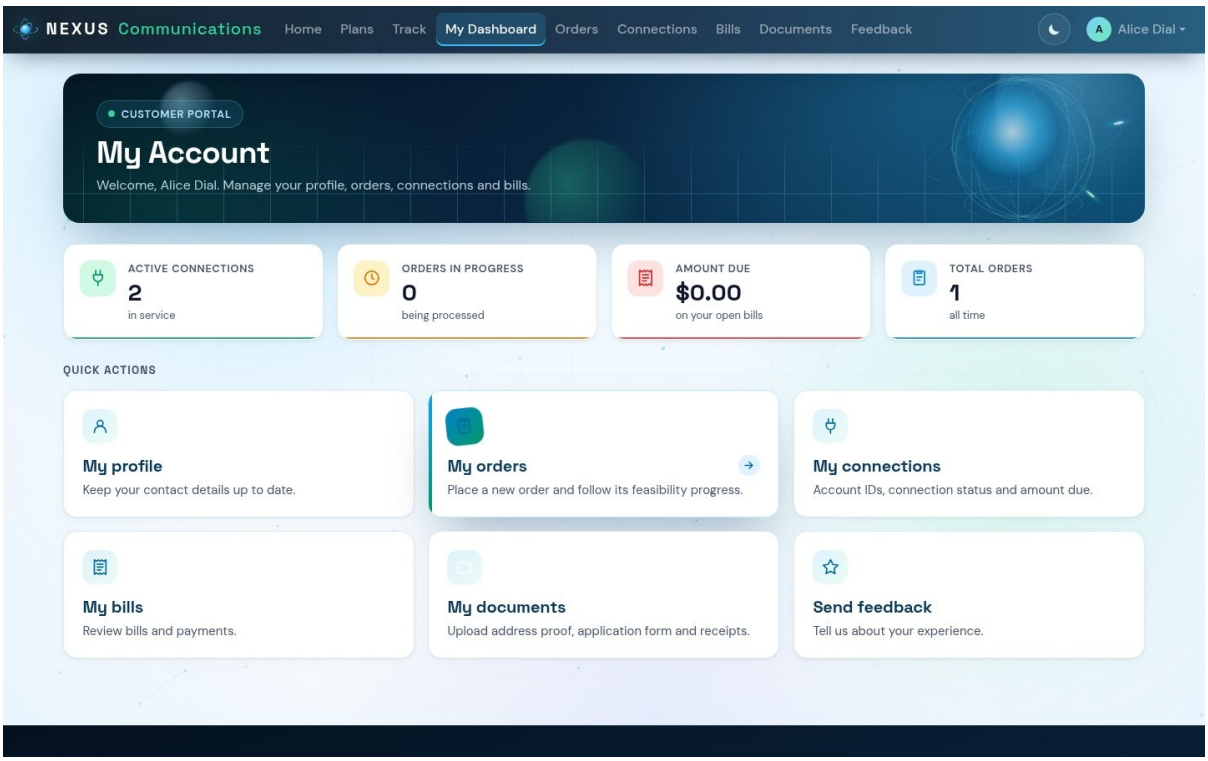


Figure 21 - Customer portal dashboard

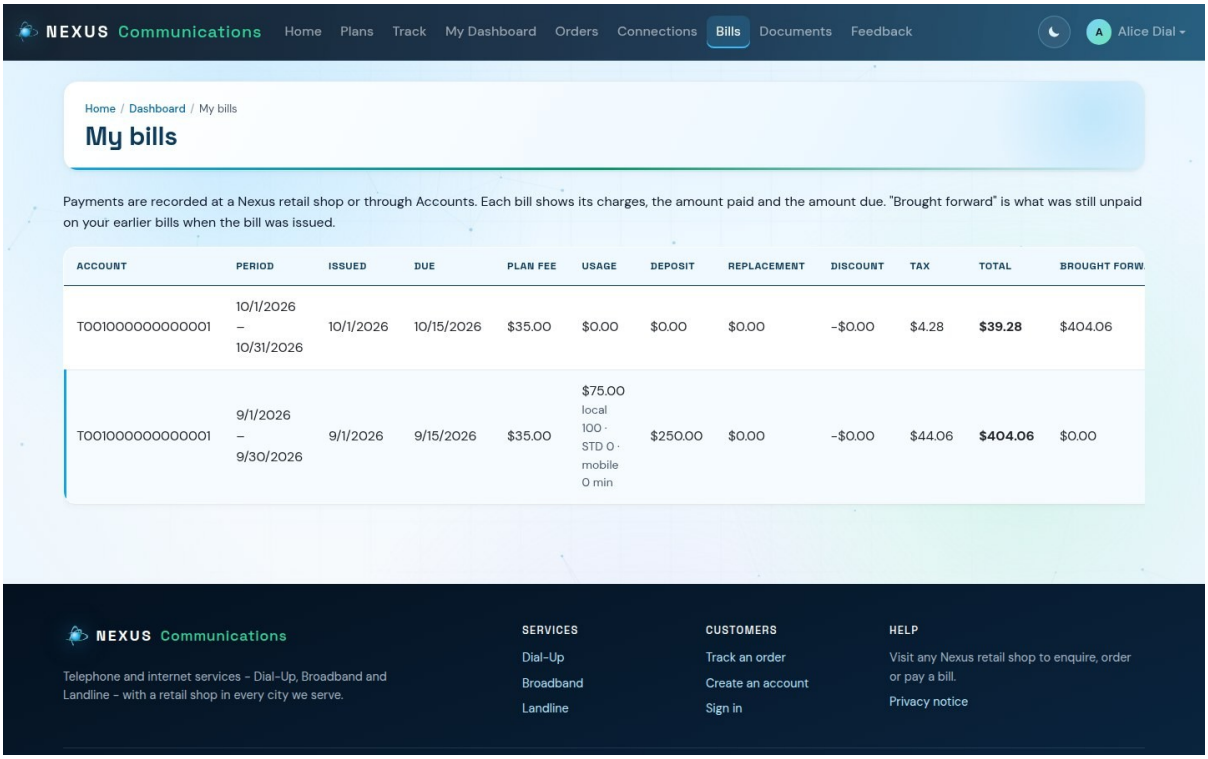


Figure 22 - Customer - itemised bills and balance

8. Login Credentials and How to Run

8.1 Requirements

- .NET 8 SDK (dotnet --version shows 8.x). Visual Studio 2022 is optional.
- SQL Server 2022 Express / Developer, or LocalDB. The database is created automatically on the first run.

8.2 Steps

1. Extract the submission ZIP (for example into Downloads or C:\). The code is in Nexus (Source Code)\Nexus.
2. Open Nexus (Source Code)\Nexus\appsettings.json and set the SQL Server name in DefaultConnection (default .\SQLEXPRESS).
3. Optional: restore database backup file\NexusServiceMarketingDb.bak in SSMS to get the demo data and logins.
4. Double-click Nexus (Source Code)\RUN-NEXUS.bat - or open NexusServiceMarketingSystem.sln in Visual Studio 2022 and press F5.
5. Open http://localhost:5294 in the browser (first start takes 1-2 minutes).

8.3 Login credentials

Role	Username	Password	Notes
Admin	admin	Admin@123	Created automatically on the first run
Retail staff	retail1	Passw0rd!x	In the database backup (Karachi shop)
Technical staff	tech1	Passw0rd!x	In the database backup
Accounts staff	acct1	Passw0rd!x	In the database backup
Customer	alice	Passw0rd!x	In the database backup; new customers use Register

With an empty database only the admin login exists; staff are added in Admin > Employees and customers register themselves.

9. Task Sheet

#	Task	Module	Assigned to	Status
1	Requirement analysis and use cases	Analysis	Ali Jan	Completed
2	ER diagram and database design	Design	Humaiz Khan	Completed
3	Entities, configurations and migrations	Database	Humaiz Khan	Completed
4	Login, roles and password hashing	Security	Muhammad Ehtisham Shakeel	Completed
5	Admin master data and stock	Admin	Ali Jan	Completed
6	Orders, bulk discount, bundled landline	Retail / Customer	Muhammad Ehtisham Shakeel	Completed
7	Feasibility and connection provisioning	Technical	Muhammad Ehtisham Shakeel	Completed
8	Billing, payments, overdue suspension	Accounts	Muhammad Waleed	Completed
9	Tracking and advanced search	Public / Staff	Muhammad Waleed	Completed
10	Customer documents and feedback	Customer	Ali Turab	Completed
11	Design system, light/dark themes, illustrations, dashboards	Interface	Ali Turab	Completed
12	Testing and documentation	Quality	All members	Completed

10. Project Review and Monitoring Report

Review point	Finding	Action taken
Combined telephone + dial-up applications	Orders could pass both checks but never be connected	Landline plan on the order; provisioning creates the line and links the dial-up
Billing completeness	Old dues and plan validity were not considered	Balance brought forward; plan fee charged once per validity period
Charges calculation	Usage was typed in by hand	Call charges calculated from minutes and plan rates
Replacement charges	Could be forgotten on the next bill	Recorded with the replacement and billed automatically once
Postpaid status	Unpaid bills did not affect connections	Technical > Overdue: suspend and reactivate
Document upload	Stored files lost their first bytes	Fixed; downloads are byte-identical
Coding standard	Some controllers had no comments	Every controller and service commented
Interface	Plain template look; too much white	Design system, light/dark theme switch, original illustrations, dashboards with live figures

11. Unit Testing Checklist

The checks below were run automatically against the running application on SQL Server 2022 (48 end-to-end checks plus browser checks). All passed.

#	Module	Test case	Expected result	Result
1	Orders	Dial-up order without a landline plan	Rejected with a clear message	**Pass**
2	Orders	Combined telephone + dial-up order	Landline plan stored; both feasibility checks created	**Pass**
3	Orders	Bulk order: second order of 6 (12 connections)	25% bulk band applied	**Pass**
4	Feasibility and connections	All checks feasible	Order becomes Feasible	**Pass**
5	Feasibility and connections	Provision combined order without a phone number	Rejected - number required	**Pass**
6	Feasibility and connections	Provision combined order	Telephone line + dial-up created and linked; 16-character account IDs	**Pass**
7	Feasibility and connections	Deposits	Landline $250, Dial-up $325; 25% band gives $187.50	**Pass**
8	Feasibility and connections	6 telephone lines with one number	Rejected - one number per line	**Pass**
9	Billing	Bill preview	Shows plan rate, replacement charges and brought-forward balance	**Pass**
10	Billing	First landline bill, 100 local minutes	35 + 75 + 250 = 360; tax 44.06; total 404.06	**Pass**
11	Billing	Call minutes on an internet connection	Rejected	**Pass**
12	Billing	STD minutes on a local-only plan	Rejected	**Pass**
13	Billing	Second monthly bill	Monthly fee charged again; 404.06 brought forward	**Pass**
14	Billing	Quarterly plan billed monthly	Plan fee only in June and September	**Pass**
15	Billing	Equipment replacement	40.00 charged on the next bill only, then marked billed	**Pass**
16	Payments and status	Overdue page	Lists only active connections with overdue bills	**Pass**
17	Payments and status	Suspend all	Overdue connections temporarily inactive; others unchanged	**Pass**

#	Module	Test case	Expected result	Result
18	Payments and status	Retail collects all bills	Bills Paid; connection offered for reactivation	**Pass**
19	Payments and status	Reactivate	Connection Active	**Pass**
20	Payments and status	Other shop opens a bill's payment form	Not found (shop isolation)	**Pass**
21	Documents, pages and interface	Upload and download a PDF	Byte-identical file returned	**Pass**
22	Documents, pages and interface	All role pages	Every page renders without errors	**Pass**
23	Documents, pages and interface	Phone width 390 px	No horizontal scrolling	**Pass**
24	Documents, pages and interface	Reduced motion	No loader; all content visible immediately	**Pass**
25	Documents, pages and interface	Theme switch (sun / moon button)	Page changes between light and dark at once	**Pass**
26	Documents, pages and interface	Theme remembered	Chosen theme stays after moving to another page	**Pass**
27	Documents, pages and interface	Theme switch on a phone	Button visible inside the collapsed menu	**Pass**
28	Documents, pages and interface	Illustrations in both themes	Drawings re-colour and stay readable	**Pass**

12. Final Checklist

Item	Status
All requirements of the problem statement implemented (chapter 2)	Yes
Separate secure login for every role	Yes
Database scripts: EF migrations and Database\NexusServiceMarketingDb-schema.sql	Yes
Every code block commented	Yes
Unit testing checklist completed - all tests pass	Yes
Interface tested on phone, tablet and desktop widths	Yes
Project report, presentation, status report and feedback form included	Yes
Team names and student IDs on every document	Yes
Database backup (.bak) added to the submission folder	Yes
Submission folder in the required structure (backup, documentation, source code, video, readme)	Yes

Appendix - Plan price list

Service	Plan	Price	Validity	Deposit
Dial-Up	Hourly 10 / 30 / 60 hrs	$50 / $130 / $260	1 / 3 / 6 months	$325
Dial-Up	Unlimited 28 Kbps monthly / quarterly	$75 / $150	1 / 3 months	$325
Dial-Up	Unlimited 56 Kbps monthly / quarterly	$100 / $180	1 / 3 months	$325
Broadband	Hourly 30 / 60 hrs	$175 / $315	1 / 6 months	$500
Broadband	Unlimited 64 Kbps monthly / quarterly	$225 / $400	1 / 3 months	$500
Broadband	Unlimited 128 Kbps monthly / quarterly	$350 / $445	1 / 3 months	$500
Landline	Local: yearly $75 (0.55/min), monthly $35 (0.75/min)	$75 / $35	12 / 1 months	$250
Landline	STD monthly $125, half-yearly $420 (rates per minute)	$125 / $420	1 / 6 months	$250